Image to Animation

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1. Function Introduction

This example program can generate a 5-second animation from a single image.

2. Hardware Connection and Setup

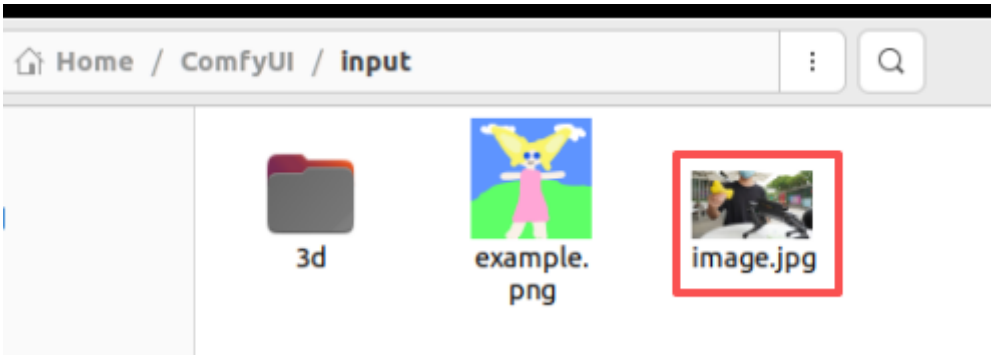
This example program requires not only a power adapter but also a DP monitor, mouse, and keyboard for displaying the interface and performing operations, or for logging into a remote desktop.

3. Code Analysis

Source code path:

```
/home/jetson/showing/animation/animation.py
```

If you need to replace the image, please replace the image in the path ~/ComfyUI/input/image.jpg.



If you need to modify the prompt message, you can modify the PROMPT variable.

```
TEXT_PROMPT = '''The background is a cyberpunk-style Tang Dynasty Never Sleeping
City, with flying eaves and holographic lanterns. The camera slowly zooms in and
circles around. The image shows a robotic dog, and a person is slowly waving a
yellow bone in front of it. The robotic dog follows the bone's sway, and the
movements are natural and seamless, with a smooth transition between real and
virtual elements. \nConstraint:\nDo not change the robotic dog's appearance.'''
```

If you need to change the output file name, modify the OUTPUT_NAME variable. The final output will be in the ~/ComfyUI/output/video/ directory.

```
OUTPUT_NAME = "myvideo"
```

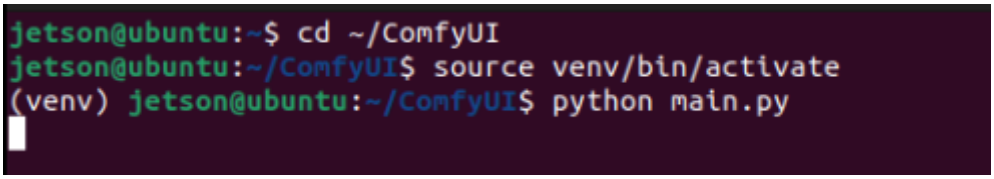
If you need to change the ComfyUI port, modify the value of the self.comfyui_server variable.

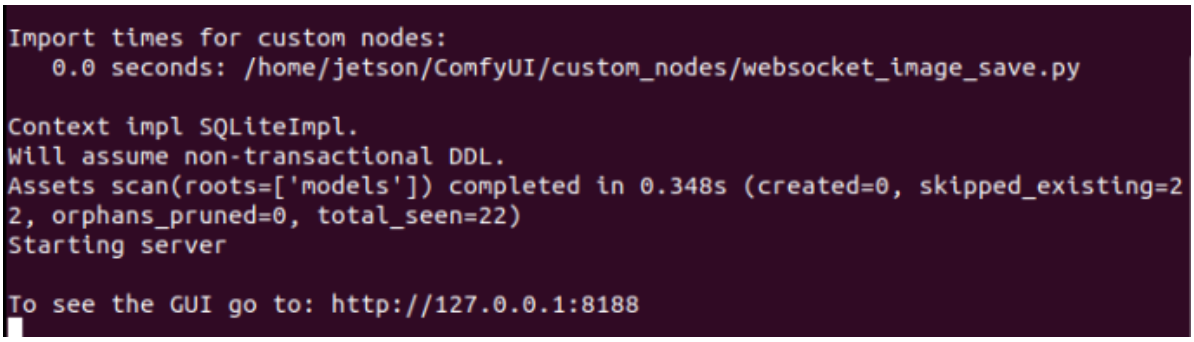
```
self.comfyui_server = '127.0.0.1:8188'
```

4. Running the code

Open a new system terminal and start the ComfyUI service.

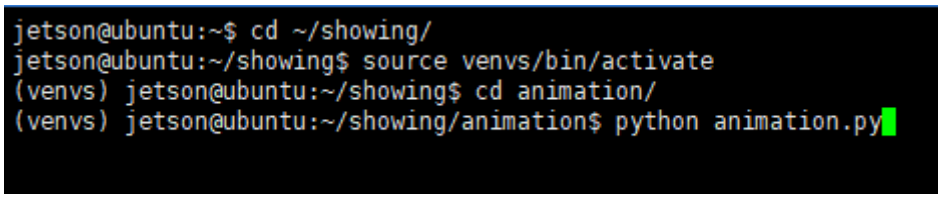
```
cd ~/ComfyUI
source venv/bin/activate
python main.py
```





Next, open a new system terminal, activate the virtual environment, and launch the main storytelling program.

```
cd ~/showing/
source venvs/bin/activate
cd animation/
python animation.py
```



5. Experimental Results

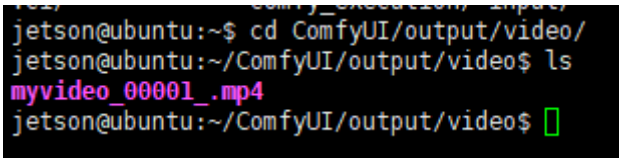
Once the program starts, it will invoke the large model files to render the animation. The execution process takes some time; with the default configuration, it requires approximately 8 minutes.

```
jetson@ubuntu: ~/ComfyUI
bj.bias']
VAE load device: cuda:0, offload device: cpu, dtype: torch.float16
Requested to load CLIPVisionModelProjection
loaded completely; 52354.41 MB usable, 787.72 MB loaded, full load: True
Found quantization metadata version 1
Using MixedPrecisionOps for text encoder
Requested to load QwenImageTEModel_
loaded completely;  8328.59 MB loaded, full load: True
CLIP/text encoder model load device: cuda:0, offload device: cpu, current: cuda:
0, dtype: torch.float16
Requested to load AutoencodingEngine
loaded completely; 35603.63 MB usable, 2408.48 MB loaded, full load: True
model weight dtype torch.float16, manual cast: None
model_type FLOW
EasyCache enabled - threshold: 0.2, start_percent: 0.15, end_percent: 0.95
Requested to load HunyuanVideo15
loaded completely; 24005.73 MB usable, 15881.76 MB loaded, full load: True
  0%|                                                    | 0/20 [00:00<?, ?it/s]
Interrupting prompt e636ec9f-87e1-4160-b0d0-ff492f90f620
  0%|                                                    | 0/20 [00:30<?, ?it/s]
EasyCache - skipped 0/20 steps (1.00x speedup).
Processing interrupted
Prompt executed in 65.73 seconds
```

```
jetson@ubuntu: ~/showing/animation
==================================================
The program has been started.
Please open your browser and log in to the ComfyUI backend (http://127.0.0.1:818
3) to check the progress.
After the video is completed, it will be automatically saved in the ComfyUI/outp
ut/video/ folder.

(venvs) jetson@ubuntu:~/showing/animation$ python animation.py
Starting Interactive Animation...
[ANIM] Generating: 背景是赛博朋克大唐不夜城，飞檐斗拱配全息灯笼，镜头慢推环绕，
图片中是一只机器狗，有一个人拿着黄色的骨头在机器狗面前缓慢摇晃，机器狗跟着黄色骨
头摇晃，机器狗动作衔接自然无卡顿，画面虚实结合，过渡丝滑无断层。
限制条件：
不要改变机器狗的样貌


==================================================
The program has been started.
Please open your browser and log in to the ComfyUI backend (http://127.0.0.1:818
3) to check the progress.
After the video is completed, it will be automatically saved in the ComfyUI/outp
ut/video/ folder.

(venvs) jetson@ubuntu:~/showing/animation$
```

Monitor the ComfyUI terminal; when the progress bar reaches 100%, it indicates that the process is complete.

```
got prompt
EasyCache enabled - threshold: 0.2, start_percent: 0.15, end_percent: 0.95
100%|                                                    | 20/20 [16:03<00:00, 48.19s/it]
EasyCache - skipped 9/20 steps (1.82x speedup).
Prompt executed in 00:17:22
```

After the animation is generated, the resulting MP4 video file is stored in the `~/ComfyUI/output/video/` directory.

```
cd ~/ComfyUI/output/video/
ls
```